1.6 Common Multiples

Let us now write the multiples of 5 and 7

Multiples of 5 are 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70,...

Multiples of 7 are 7, 14, 21, 28, 35, 42, 49, 56, 63, 70,...

Here, the common multiples of 5 and 7 are 35 and 70 and will go on without ending.

As multiples of a number are infinite, we can think of the **Least Common Multiple** of numbers, shortly denoted as **LCM**.

1.6.1 Least Common Multiple (LCM)

Think about the situation:

Situation 1: Write the multiplication table of 4 and 5 (upto 10).

4th Table	5th Table
1x4 = 4	1x5 = 5
2x4 = 8	2x5 = 10
3x4 = 12	3x5 = 15
4x4 = 16	4x5 = **20**
5x4 = **20**	5x5 = 25
6x4 = 24	6x5 = 30
7x4 = 28	7x5 = 35
8x4 = 32	8x5 = **40**
9x4 = 36	9x5 = 45
10x4 = **40**	10x5 = 50

Observing the multiplication tables, can you find the multiples (product of numbers) that are the same in the 4th table and 5th table?. If yes, what are they? Yes, they are **20, 40,**...etc. From the multiples of 4 and 5, we can easily find that 20 is the least common multiple of 4 and 5.

Situation 2:

Anu wants to buy Ragi Laddus and Thattais to serve at her sister's birthday party. Ragi Laddus come in packets of 4 and Thattais come in packets of 6. Anu has to buy these packets so that there are the same number of Ragi Laddus and Thattais to serve at the party. How will Anu tackle this situation?

This situation can be tackled by Anu using the concept of LCM. Here, multiples of 4 are 4, 8, 12, 16, 20, 24, ... and multiples of 6 are 6, 12, 18, 24, 30, 36, ... We find that the common multiples are 12, 24, ... of which 12 is the least common multiple. Hence, Anu should buy a minimum of 3 packets of Ragi Laddus and 2 packets of Thattais so that there are the same number of Ragi Laddus (12) and Thattais (12) to serve at the party.





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